Algorithmic Trading: COMP0051

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I INTRODUCTION:

This report compares the effectiveness of two algorithmic trading strategies using the SPDR® Portfolio Long Term Treasury ETF (SPTL). The strategies are analyzed against the U.S. monetary policy, represented by the Effective Federal Funds Rate (EFFR). This paper evaluates the alpha-generating capabilities of leveraged trading strategies relative to a risk-free investment benchmark. The SPTL ETF tracks U.S. Treasury bonds with maturities exceeding 10 years. The EFFR, provided by the Federal Reserve Bank of New York, denotes the interest rate at which financial institutions lend reserve balances to each other overnight, serving as the risk-free rate. The paper first prepares time series data for SPTL and EFFR, then designs two trading strategies, evaluates their performance using key indicators, and draws a conclusion based on the results and discussion.

II DATA PROCESSING & TIME SERIES PREPARATION

Data for the SPTL ETF was sourced from Yahoo Finance [1], and the EFFR from the Federal Reserve Bank of New York [2]. Both datasets cover the entire trading period from January 1st to December 31st, 2023, consisting of 252 trading days. The risk-free rate (EFFR) was converted from an annual rate to a daily rate using a day-count approximation (dc $\approx \frac{1}{252}$). The daily excess return per unit of SPTL was computed using backward differencing where the change in price for SPTL was calculated between time step t and $t-1$: $r_t^e = \frac{\Delta p_t}{p_t} - r_t^f$. Below are two plots, Figure 1 observes the daily returns of EFFR and Figure 2 shows the excess return rate.

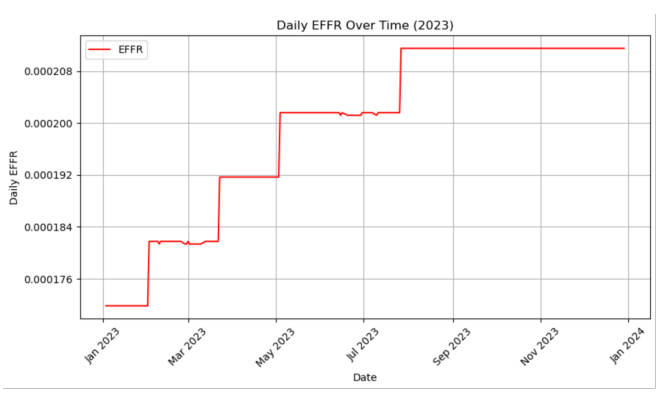

Figure 1: Observes the EFFR consistently increasing throughout 2023 due to restrictive monetary policy by the Federal Reserve, driven by inflationary pressures and macroeconomic factors such as labor market conditions and persistent inflation [3].

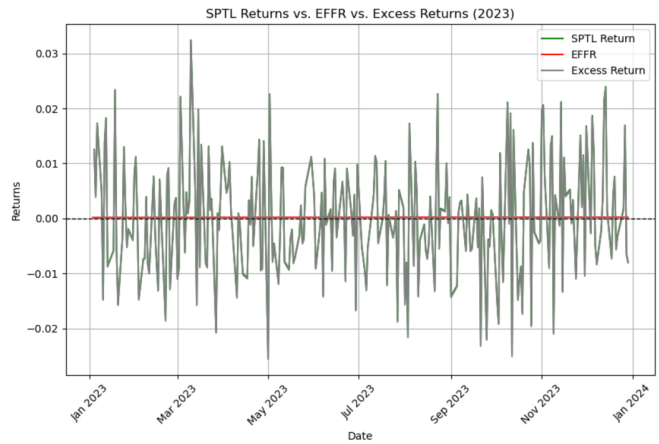

Figure 2: This timeframe allows for a consistent day-by-day comparison of SPTL returns (green), EFFR interest rates (red) and excess return (grey). The impact of the interest rates are close to zero when comaparing to the SPTL return. The data serves as the basis for constructing and evaluating two leveraged trading strategies.

III METHODOLOGY

This paper considers two quantitative trading methods: a 15-day Lookback Momentum strategy and a Trend-Following approach using moving averages (MAs). To optimize returns and minimize risks multiple trading strategies were tested with historical data. Of these strategies, two leveraged strategies are used to evaluate their ability to capitalize on market movements and generate returns. To ensure that the data solely relies on historical data available at the time of making decisions, this paper uses backward differencing. The condition for the strategies where an initial capital (V_0) of \$100,000, leverage factor (L) of 10, implying a maximum initial exposure of \$1 million and unused capital is put in a money-market growing at the EFFR risk-free rate. For both strategies the trading capital remained fixed at \$100,000, with exposure adjusted daily based on previous day's signals.

If profits occurred, they were segregated into the money-market account. Losses directly reduced trading capital. In cases where trading capital fell below \$50,000, funds from the money-market account replenished it to a maximum of \$100,000.

The performance of the trading profit or loss (PnL) is calculated by taking the product of excess returns and exposure held

$$\Delta V_t = \frac{\Delta p_t}{p_t} - r_t^f \, \theta_t.$$

then it captures interest on unused capital in a money-market account

$$\Delta V_t^{\text{cap}} = V_t^{\text{total}} \, r_t^f - M_t \, r_t^f, \quad \text{with} \quad M_t = \frac{|\theta_t|}{L}.$$

Finally, it combines these to calculate the total daily change in account value

$$\Delta V_t^{\text{total}} = \Delta V_t + \Delta V_t^{\text{cap}}.$$

plotting each component and their cumulative values.

i Strategy 1: 15-Day Lookback Momentum Strategy

As a first strategy, this paper uses a 15-day Lookback Momentum strategy to compare recent price levels with historical price levels and to capture intermediate-term trends. For this strategy a daily mark-to-market approach is applied. The open trading position is evaluated at the end of each trading day, and the profit or loss realized for that day is immediately reflected by transferring gains into the money market. This means that the portfolio's performance is updated daily.
Since the annual volatility of the SPTL ETF is 17%, indicating an average volatility, this strategy employed a 15-day lookback period, thereby balancing the need for timely signals with lower transaction costs [4]. Using a 15-day threshold for buy and sell signals, preserves responsiveness to quick changes in trends while smoothing over day-to-day noise. The lookback window is used to determine if today's price is higher or lower than the price from 15 days prior, generating buy (+1) signals when positive momentum is detected and sell/short (-1) signals when the momentum is negative:

$$s_t = \begin{cases} +1, & \text{if } p_t > p_{t-15}, \\ -1, & \text{if } p_t < p_{t-15}, \\ 0, & \text{otherwise (i.e., } p_t = p_{t-15}). \end{cases}$$

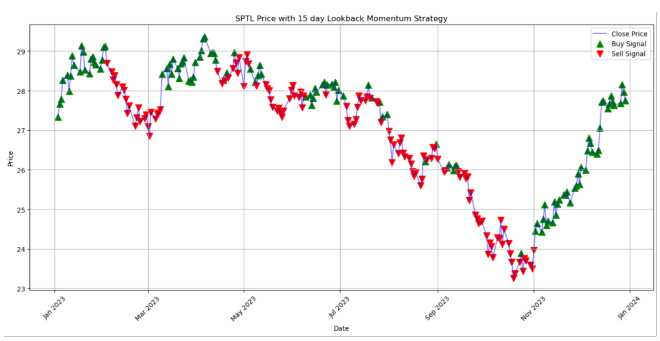

Figure 3: Illustrates the daily trading signals based on the 15-day lookback period.

ii Strategy 2: Trend-Following

Next, this paper analyzes a Trend-Following approach that exploits medium- to longer-term price direction using two MAs. Different to the first strategy, this strategy triggers a trade only when a fast-over-slow crossover occurs, but the open position is still revalued each day. That is, a daily trading PnL is computed and the corresponding gains are transferred between the trading capital and the money market every day.
A fast MA of 20 days and a slow MA of 50 days are defined. A buy (+1) signal is generated when the fast MA crosses above the slow MA, indicating an upward trend, while a sell/short (1) signal is generated when the fast MA crosses below the slow MA. This event-based approach helps filter out random noise and avoids excessive trading, while still capturing critical turning points in price.

$$s_t = \begin{cases} +1, & \text{if } MA_{\text{fast}}(t) > MA_{\text{slow}}(t), \\ -1, & \text{if } MA_{\text{fast}}(t) < MA_{\text{slow}}(t), \\ 0, & \text{otherwise.} \end{cases}$$

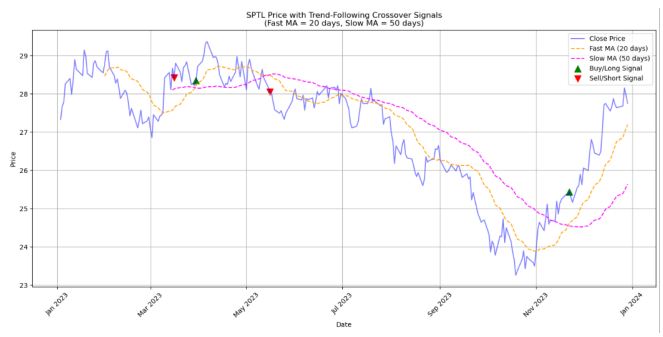

Figure 4: This graph shows all signals that are triggered when a MA crossover occurs. The first signal is triggered after 50 days (length of the slow MA).

iii Performance Indicators

To measure the risk adjusted return for both profit levels and risk exposure, two commonly used performance metrics, the Sharpe and Calmar ratio are used. The Sharpe ratio focuses on the volatility of returns relative to the risk-free benchmark, while the Calmar ratio emphasizes drawdowns and highlights how effectively a strategy avoids severe capital losses. Using the total return for the two strategies ($r_t = \frac{\Delta V_t^{\text{total}}}{V_t^{\text{total}}}$) the Sharpe and Calmar ratio are calculated.
Sharpe Ratio: Measuring risk-adjusted returns compared to volatility.

$$\text{Sharpe Ratio} = \frac{E[\%PnL_t]}{\sigma[\%PnL_t]}.$$

Calmar Ratio: Annualized return relative to the maximum drawdown, assessing long-term effectiveness against downside risk.

$$\text{Calmar Ratio} = \frac{\text{Annualized \% Return}}{\text{Maximum Drawdown}}.$$

IV RESULTS:

In Figure 5, both the Lookback Momentum approach and the Trend-Following method are evaluated against a passive Buy-and-Hold benchmark. With a starting capital of $100,000 and employing 10× leverage, the Buy-and-Hold strategy adjusted to the risk-free-rate reached a final portfolio value of $93,000, translating to a –7.44% total return. In contrast, the Lookback Momentum strategy exploited the directional price patterns more aggressively. The Lookback Momentum strategy concentrates on recent performance signals switching between long and short positions, while the Trend-Following model less successful aims to capture directional moves. When a long-term trend is observed, such as when the SPTL ETF price decreases from July – November, the Lookback Momentum strategy performs much better than both of the other strategies.

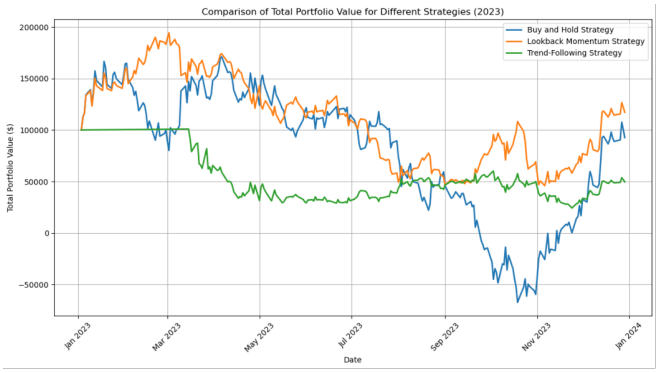

Figure 5: The plot illustrates the total portfolio value for the Momentum strategy (orange), the Trend-Following Strategy (green) and the Buy and Hold Strategy (blue).

i Strategy Analysis

The inherently reactive nature of the Lookback Momentum strategy identifying short-term trends in market behavior helps the strategy pivot quickly after strong market changes. Because it uses 10× leverage, a small net price movements are magnified, meaning that the strategy can produce significant gains if the SPTL ETF continue moving in the anticipated direction. Nonetheless, this strong dependance on recent price patterns also pose challenges. Should the market experience periods where no clear trend emerges or reversals happen too fast, momentum signals may lead to frequent position changes that decrease returns. This challenge is seen in Figure 6 around the time period of March 2023, where quick reversals lead to a drop in the total portfolio value. Therefore, the strategy demands consistent monitoring to ensure that the short-term trend component remains aligned with prevailing price movements.

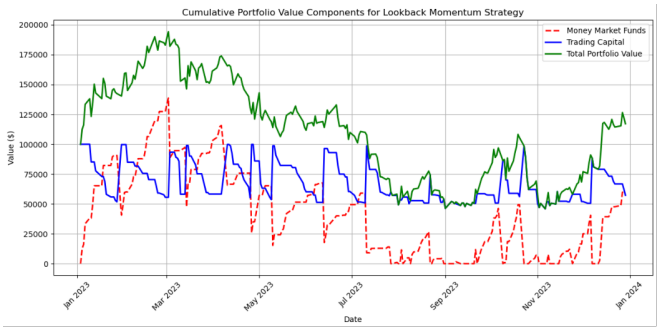

Figure 6: Shows the strategy's total portfolio value components over time. Notably, the strategy ends with a positive cumulative return of 17.14% and a final portfolio value of $117k.

The Trend-Following strategy exploit sustained price direction, focusing on whether the market is predominantly rising or falling over a longer window. By emphasizing extended moves rather than short-term fluctuations, it generates fewer trades than a pure momentum approach. When combined with a 10× leverage factor, positions can grow significantly as the trend endures, converting gradual gains into outsized profits. However, in the 2023 data, a lengthy sideways market as well as abrupt shifts in price direction negatively impacts the Trend-Following strategy, since signals may lag until a clear path emerges [5]. This delay is observed in Figure 7, as in the beginning up until July 2023 the SPTL EFT prices move without a clear upward or downward trend leading to greater losses and minimal growth. Overall, the Trend-Following framework, remains a valuable mechanism for capturing prolonged directional price action.

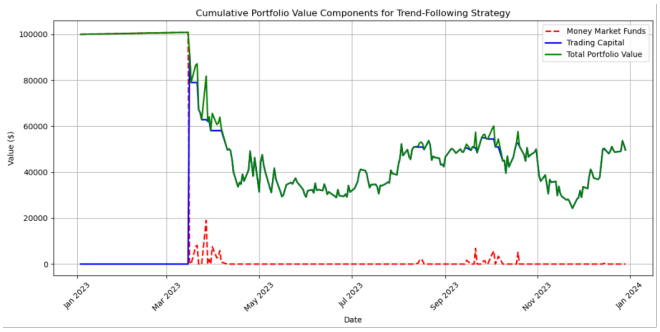

Figure 7: Shows the strategy's total portfolio value components over time. The strategy ends with a negative cumulative return of -50.35% and a final portfolio value of $50k.

ii Performance Evaluation

In Figure 8, the Sharpe ratio and the Calmar ratio are compared for the Lookback Momentum and Trend-Following strategies, using the total daily returns of each strategy. These risk-adjusted metrics reveal how each approach performs relative to volatility (Sharpe ratio) and the depth of drawdowns (Calmar ratio). The sample size and external market factors, such as continuously rising

EFFR rates, affect the reliability and magnitude of these results.

As seen in Figure 8, the Lookback Momentum delivered a 17% total return, supported by a Sharpe Ratio of 0.75, indicating solid risk-adjusted performance despite market volatility. Its Calmar Ratio of 0.22 highlights effective drawdown control, as temporary fluctuations only had limited impact on the overall gains. The strategy's shorter-term focus allows it to capitalize on rapid price movements, particularly in shifting rate environments, while mitigating losses [4].

Trend-Following achieved a higher Sharpe Ratio (1.28), reflecting efficient risk-adjusted returns, but suffered a -50% total loss and a negative Calmar Ratio (-0.66), revealing big drawdowns relative to performance. Its longer MA crossover reduces reaction to short-term noise, resulting in fewer but more deliberate trades. While this improves Sharpe metrics, the strategy struggles to recover from longer downturns.

The Buy & Hold benchmark posted a -7% return, with a Sharpe Ratio of 1.68 (the highest) but a slightly negative Calmar Ratio (-0.05), underscoring its vulnerability to extended declines despite smoother volatility-adjusted returns.

While Trend-Following excels in risk efficiency (Sharpe), its deep drawdowns make it less reliable outright. Lookback Momentum strikes a balance, offering stronger absolute returns and drawdown resilience, particularly in rising-rate regimes [6]. Each approach presents distinct trade-offs, but the Lookback Momentum's stability performas stronger in uncertain markets.

Performance Indicators

Metric	Lookback Momentum	Trend Following	Buy & Hold Benchmark
Total Return	17%	- 50%	- 7%
Sharpe Ratio	0.75	1.28	1.68
Calmar Ratio	0.22	- 0.66	- 0.05

Figure 8: Compares the total return, sharpe and calmar ratio with the three discussed strategies.

V DISCUSSION

The results indicate that the Lookback Momentum and Trend-Following strategies each have distinct strengths and drawbacks when applied to the SPTL ETF under rising interest rates. The Momentum strategy achieves a substantially positive return for the 2023 period, capitalizing on short-term market swings and quickly reversing positions in response to directional shifts. Profits from these trades were put aside in a money-market account, building a reserve with the intention to protect the gains from later fluctuations. This quick reactions also means it's easily affected by frequent price changes during times of high market volatility. As seen in the figures, rapid shifts during mid-year led to consecutive position changes that undermined performance, with losses directly reducing trading capital and testing the strategy's resilience. Nevertheless, the combination of a shorter lookback window and 10× leverage proved effective overall, particularly when consistent directional moves emerged in the latter half of 2023.

In contrast, the Trend-Following strategy demonstrates mixed outcomes, as its reliance on longer MA crossovers weakens its ability to respond swiftly to short-lived reversals. This approach results in fewer trades and less unnecessary buying and selling, but in the current market conditions, signals often came too late or too early. The strategy's deeper drawdowns triggered the capital-replenishment rule multiple times, with funds from the money-market account restoring trading capital as it fell consistently below $50,000. While this mechanism intended to prevent total depletion, the repeated withdrawals from the money-market reserve dissolved the buffer needed to sustain recovery during prolonged downturns. Therefore in this case, the restoring trading capital mechanism was not helpful, concluding with a lower terminal value than both Lookback Momentum and the Buy-and-Hold benchmark.

Several factors help explain these results. The 10× leverage amplified both profits and losses, with the money-market segregation only mitigating risks for the Lookback Momentum strategy but not for the Trend-Following strategy as it was exposed to the compounding pressure of drawdowns. The steady rise in interest rates made borrowing more expensive, cutting into profits from leveraged trades. While this helped returns on unused cash reserves, it wasn't enough to fully offset the added costs.

VI CONCLUSION

In summary, the Lookback Momentum strategy shows strength in reacting to short-term price changes, delivering steady performance even in a year shaped by rising interest rates. The Trend-Following approach, while generally effective in clear, long-lasting trends, struggled in this setting due to slower signal timing and long stretches of flat or unpredictable movement. Both strategies demonstrate how using leverage can increase swings in performance and risk, pointing to the potential benefits of more flexible position sizing to help protect capital.

Looking ahead, improvements like adjusting leverage based on market conditions and factoring in trading costs could help address some of the challenges. Also expanding the timeframe to include dedicated train-test splits would validate the model's robustness across diverse market conditions, improving its applicability while reducing overfitting risks. In general, it is important to be mindful of the limited data, shifting interest rate policies, and the risks that come with high leverage.

References

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